BankShield: Fairness-Aware Safety Guardrails for Indian-Language Banking AI

Piyush Agarwal
Independent Researcher, India
mynameispiyush@gmail.com

Abstract

Safety guards for Indian-language banking must detect discrimination without changing verdicts solely because a demographic identity changes. We introduce GUARDAUDIT, an evaluation protocol combining decontamination, paired harmful/benign calibration, counterfactual identity tests, and deployed-action evaluation, plus two released banking benchmarks. Across six guards, the hardest matched test remains unresolved: the best system flags a coded proxy denial while clearing its matched actuarial control on only 37.5% of pairs. BankShield, used as a case study, reaches 20.0% and false-flags 75.0% of expert-validated legitimate controls. On separate BharatBBQ subsets, it achieves 0% flip on benign identity substitutions and 66.7% biased-content catch; these disjoint measurements do not establish joint per-example fairness. Its unadapted 12B backbone supplies near-invariance and attack recall, while adaptation and a discrimination probe raise biased-content catch from 24% to 43% and 66.7%, not significantly above a smaller Indic guard's 56.9% (p=0.09). Detection also falls from 81.6% in English to 42.6% in Devanagari Hindi. BankShield is therefore a diagnostic case study, not a deployment-ready fairness solution. We release the protocol, benchmarks, and frozen predictions.

1 Introduction

Conversational AI is increasingly deployed in Indian retail banking. Banks serve large numbers of customers through messaging-platform assistants in regional languages, increasingly mediated by LLMs that not merely answer but act: initiating transfers, updating KYC records, triggering account actions. The safety layer sits between a customer and their money, so an error is not a rude sentence but a financial event. A wrongly blocked request denies a customer their own account; a wrongly allowed one advances a fraud. Those most exposed are the first-time and vernacular users financial inclusion is meant to reach.

The guardrails governing these systems were built for English, and their assumptions break in Indian banking on three fronts (§2). Customers write in code-mixed, romanized text, and adversarial inputs exploit exactly these surface forms that English-first filters cannot read (Aswal and Jaiswal, 2025). The banking harm surface (OTP/UPI fraud, impersonation, KYC bypass, laundering) does not resemble general toxicity, while ordinary banking terms (*foreclosure*, *lien*) are misread as unsafe. And regulation—the Reserve Bank of India's requirement that covered payment-system data be stored in India (Reserve Bank of India, 2018), its IT-outsourcing directions (Reserve Bank of India, 2023), and the DPDP Act (Government of India, Ministry of Electronics and Information Technology, 2023)—makes sending customer data to externally hosted models a compliance risk for these institutions, so banks in our target segment adopt an on-premise guardrail as an operational requirement. Beyond detecting harm, such a system must also treat customers equitably across demographic identities, a requirement existing banking safety evaluations rarely examine, and the central concern of this paper.

This reframes the problem: there is no escape hatch. Escalating a misclassified query to a larger hosted model would conflict with these institutions' on-premise data requirements, so multilingual safety shifts from *coverage* (more languages) to *sufficiency* (whether a compliant self-hosted system is good enough alone), and must be evaluated under the on-premise constraint this target deployment imposes.

Recent work addresses Indic safety, establishing that a dedicated Indic guard outperforms multilingual baselines; in financial NLP, domain-specialized models trained on regulation-grounded data outperform general baselines on compliance tasks (Jiang et al., 2025; Wang et al., 2025). But this work targets regulatory *knowledge* and broad social harm, is trained largely on machine-translated data (stripping the code-mixing real banking attacks use), and is evaluated on held-out sets rather than in deployment. The missing question is whether the guard itself behaves fairly: whether its verdict changes when only a demographic attribute (caste, religion, region, gender) varies. Since fine-tuning can induce bias and a biased guard silently over-polices some groups, this is not optional for a regulated deployment.

We address this with BankShield, our safety framework for Indian-language banking AI, evaluated as a compliance-aware system in a self-hosted volunteer test pilot (283 test moderation events generated by volunteer testers, with no real customers and no banking actions taken; Appendix M). Our central finding reframes what fairness means for a guard: fairness is not invariance; it is invariance without blindness, since a guard blind to a discriminatory denial is not fair to the customer denied. We argue fairness auditing should be standard in evaluating multilingual high-stakes safety guards. Our contributions:

- **GuardAudit, a reusable evaluation protocol (our primary contribution).** We consolidate the checks a high-stakes multilingual guard should pass—decontamination, paired detection/over-block with confidence intervals, a counterfactual demographic-fairness audit, and deployed-action (block/route/allow) evaluation—into GuardAudit (Appendix B), a protocol that is model-, language-, and domain-agnostic. We release it with evaluation and scoring code, prompts, and frozen per-item predictions.[1]

- **Two banking benchmarks and a structured failure taxonomy.** Grounding fairness in banking decisions (RBI Charter grounds; coded-discrimination patterns from Indian case law) and adding a direction-aware output-safety benchmark, we find (i) no guard reliably separates coded proxy-discrimination from legitimate adverse decisions (best paired accuracy 37.5%); (ii) the failures are *structured*, not random: general guards are proxy-blind (clearing both halves of a pair), whereas BankShield is the mirror image (flagging both); and (iii) safety competence is domain-specific. To our knowledge this is the first counterfactual demographic-fairness audit of Indic *banking* guards; counterfactual-fairness auditing of English safety classifiers is established (Sturman et al., 2024; Achara and Chhabra, 2025).

- **BankShield as an honestly evaluated case study.** We apply GuardAudit to BankShield, a self-hosted guard with direction-aware moderation. It leaves benign identity content unflagged (0% flip) and catches biased content (66.7%), but on the matched proxy/control test reaches only 20.0% paired accuracy and over-flags 75.0% of expert-validated legitimate controls; its 66.7% catch does not significantly exceed a smaller Indic guard's 56.9% (p=0.09), and most of its general competence is inherited from an open backbone (biased-content catch 24% $\rightarrow$ 43% $\rightarrow$ 66.7% is what adaptation adds; §4.3). The audit finds BankShield useful for detection but not deployment-ready for autonomous discrimination-sensitive decisions (§6.3–§6.5).

Where prior work asks whether an Indic guard can be built, we ask how to *measure* whether one is fair, and contribute GuardAudit and two banking benchmarks, with BankShield as an honestly evaluated case study. The audit's verdict on BankShield is negative where it matters most: on coded discriminatory denial no guard including ours separates proxy from legitimate risk, and BankShield over-flags 75.0% of expert-validated legitimate controls, so it is not ready for autonomous use on discrimination-sensitive adverse decisions. There our contribution is the measurement, not a solution.

[1] The artifact contains evaluation and scoring code, prompts, decontaminated benchmark sets with item IDs, frozen per-item prediction logs, expected metrics, environment and model-version information, the disclosable policy configuration, and a SHA-256 manifest, in a public repository released with this paper (`https://github.com/PiyushAgarwal9/bankshield-finnlp-artifacts`). Its scripts recompute the reported metrics from frozen predictions, and public baselines can be rerun. Independent end-to-end BankShield inference is not reproducible: the proprietary LoRA weights and production policy rules are withheld under the organizers' written exception, and the exact deployed discrimination-probe checkpoint is unavailable. The base model (`google/gemma-4-12B-it`) is public.

2 The Indian Banking Safety Problem

We characterize what makes safety for Indian-language banking assistants a distinct problem, and why a general-purpose guard, even one adapted to Indic languages, cannot simply be dropped into this setting.

2.1 A domain-specific harm surface, in both directions

The harms that matter in banking differ from general taxonomies. A banking assistant faces attacks whose surface form is often polite: prompt injection, fraud and social engineering (soliciting an OTP or UPI PIN, impersonating a bank officer), KYC bypass, money-laundering, none carrying toxicity markers yet exactly what a banking guard must catch. The dual failure is over-blocking: *foreclosure*, *lien*, *encumbrance* read as threatening, and code-mixing compounds this. Generic filters thus both miss real attacks and deny service on ordinary terms; no single threshold fixes this. It needs a model whose notion of harm is domain-aligned, attending to intent and directionality, not surface offensiveness.

2.2 Regulation, and why blocking is not the only action

RBI rules require covered payment-system data to be stored in India while permitting overseas processing subject to specified return, storage, and deletion conditions (Reserve Bank of India, 2018); RBI's IT-outsourcing directions require regulated entities to retain oversight, auditability, and responsibility (Reserve Bank of India, 2023), while the DPDP Act governs lawful processing of personal data (Government of India, Ministry of Electronics and Information Technology, 2023). Banks in our target segment consequently impose on-premise processing as an operational requirement; we do not present this as a universal statutory mandate. A guard sending customer messages to a cloud-hosted model thus raises outsourcing and auditability concerns for these institutions; they require on-premise deployment, which forces a self-hosted guard sufficient on its own. An RBI Bulletin speech also recommends fairness audits and explainability for financial AI (Rao, 2024). Safety here is also not binary: a customer in distress or with a grievance is not producing content to block, and refusing it is itself a harm. BankShield therefore distinguishes *allow*, *block*, and *route to human*, which matters for measurement, since counting routed grievances as over-blocks would misrepresent the system (§6).

3 Related Work

We situate our work (safety for AI deployed in Indian-language banking) against three adjacent lines, each a foundation we build on rather than the space our contribution occupies.

Guard models and safety evaluation. The prevailing approach to LLM safety is an external moderation layer returning a verdict on a prompt or response, e.g. Llama Guard (Inan et al., 2023) and ShieldGemma (Zeng et al., 2024), fine-tuned on datasets such as AEGIS (Ghosh et al., 2024) and BeaverTails (Ji et al., 2023). This literature also produced instruments we adopt, notably XSTest (Röttger et al., 2024), which isolates exaggerated safety, the over-refusal of benign but sensitive prompts. This work is overwhelmingly English, with taxonomies that do not transfer across languages (Adilazuarda et al., 2024), and reports safety as an aggregate score, whereas a guard's behaviour is visible only in the pair: harm caught and legitimate content wrongly blocked (§5).

Multilingual and culturally-aware moderation. A recent shift extends safety beyond English: for the Indic space, CultureGuard (Joshi et al., 2025) and related work establish that Indic safety is distinct (harm is socio-culturally situated) and release Indic-specialized guards, the closest prior work to ours and a foundation we build on. Recent multilingual guards (PolyGuard (Kumar et al., 2025), Qwen3Guard (Qwen Team, 2025), and the LoRA-adapted SEALGuard (Shan et al., 2025)) broaden language coverage but are neither domain-specialized nor fairness-audited. All of these, Indic and multilingual alike, are by design general-purpose guards evaluated as research artifacts: they do not target a specific domain, are not evaluated as deployed systems under regulatory constraint, and do not audit whether the guard's own verdicts are demographically biased: scope boundaries that describe the space our work occupies.

Domain-specific, deployed, and fairness-aware safety. Three further threads bound our contribution. Most central is fairness of the guard: prior Indic-safety work treats identity as a category of harm to detect (Sahoo et al., 2024; Nawale et al., 2025; Khandelwal et al., 2024), and counterfactual and perturbation-based fairness analyses exist for English toxicity classifiers (Dixon et al., 2018; Prabhakaran et al., 2019; Garg et al., 2019; Kusner et al., 2017), but no prior work asks whether an Indic guard is itself biased when only a protected attribute changes. The other two are deployment (research models do not face the latency, localization, and auditability of a live banking system) and domain: FinGuard (Dou et al., 2026) targets regulatory-compliance violations, not the guard's own fairness, and covers no Indic language. Financial safety is beginning to receive dedicated benchmarks (FinSafetyBench (Hou et al., 2026), evaluating LLM safety across financial scenarios) and deployed systems (Banking Done Right (Chua et al., 2025), a language-centric retail-banking assistant), but neither is Indic, code-mixed, nor a counterfactual fairness audit of a guard's own verdicts; the Indic banking harm surface and the fairness of the guard governing it remain open.

Positioning. Prior work establishes that Indic safety matters and that a dedicated Indic guard is achievable, foundations we build on. Our contribution is not another Indic guard but GUARDAUDIT, an evaluation framework for the fairness of deployed multilingual banking safety systems, demonstrated through BankShield (Table 7).

4 BankShield

Conceptually, BankShield is a cascaded safety system mediating each turn between a customer and an assistant. A message undergoes deterministic normalization, then a multilingual classifier and lightweight probes; a configurable policy engine maps their outputs to one of three actions: allow, route to a human, or block (Figure 1, Appendix A). As a guard on regulated data we release evaluation artifacts but not production rule sets or weights, for which the workshop organizers granted a commercial/security exception in writing. We emphasize the decisions distinguishing BankShield from a general-purpose guard sharing its backbone. The architecture supports direction-aware and per-category moderation (§4.3, §4.5); our evaluation tests whether this produces acceptable fairness rather than assuming it does.

4.1 A cascaded, cost-aware pipeline

Each message passes through a short cascade. A *deterministic lexical filter* normalizes the text (inserted spacing, leetspeak, repetition) and blocks on exact lexicon match at zero false-positive cost. The classifier (§4.2) then emits a category and calibrated severity in one pass, lightweight probes (§4.4) score dimensions warranting independent re-trainable detectors, and a policy engine (§4.5) maps these to one of three actions, keeping the decision auditable; unambiguous lexical matches (rare in practice, so most traffic reaches the classifier) block at zero model cost.

4.2 Backbone and adaptation

We use an open, mid-sized backbone because it can be served on-premise while retaining multilingual capability (a requirement, not a preference) under §2.2. The classifier is a LoRA-adapted `gemma-4-12B-it`[2] in 4-bit (NF4) quantization, producing category and severity by a single deterministic pass (log-probabilities over fixed label tokens, not free-form generation); hyperparameters are in Appendix M. The general-purpose Indic guard we compare against uses a smaller, earlier `gemma-3-4b-it` backbone; to isolate the effect of adaptation rather than scale, our base-model reference in §6 is the same `gemma-3-4b-it` backbone the competitor builds on, un-adapted.

4.3 Direction-aware moderation

BankShield moderates inputs and outputs differently. The central asymmetry concerns discrimination: it detects it on the *output* side (assistant responses) but does not flag a customer's *input* as discrimination, because the customer is the protected party, not the source of harm: flagging their identity mention or grievance as a violation would punish the users the system serves. This choice about *whose* safety the guard protects underlies the fairness behaviour of §6.2.

4.4 Lightweight probes

Three probes, each a logistic model over multilingual E5 sentence embeddings (Wang et al., 2024) from an on-premise encoder (preserving the localization constraint of §2.2), retrainable in seconds and independent of the 12B model, score dimensions warranting dedicated, quickly-adjustable detectors: *grievance*, *discrimination* (output side),

[2]Gemma 4 (`google/gemma-4-12B-it`, `https://huggingface.co/google/gemma-4-12B-it`) is a public Google release; a distinct, later model than the `gemma-3-4b-it` the Indic guard uses, not a typo.

and *abuse-elicitation*; a fourth signal, the deterministic *lexical filter* (§4.1), is rule-based. Separating these from the main model lets an operator adjust a behaviour (say, grievance-routing sensitivity) without retraining the classifier or other probes.

4.5 A three-way action model

BankShield does not treat safety as binary: its policy engine maps each message to *block*, *route-to-human*, or *allow* by per-category thresholds and probe outputs. Distress and grievance signals (self-harm, complaints, consent-sensitive requests, abuse-elicitation) trigger *routing*, not a block: a customer in distress must be escalated, not refused. This also matters for measurement (§6): we count only *block* as a refusal, treating routing as designed escalation.

4.6 Per-tenant policy without retraining

The model and probes are shared, but policy is per-institution: category strengths, actions, moderation directions, probe thresholds, and denied topics are set per-tenant and tuned to a bank's risk appetite without retraining, letting one deployed system serve institutions with differing compliance requirements.

4.7 Training-data stance

Driven by the privacy and localization constraints of §2.2, BankShield is trained without machine translation and without real customer data. Its ≈172K-example corpus (51% safe, 49% unsafe) is 99.6% synthetic, with the remaining 0.4% (686 items) drawn from public benchmarks; all are authored directly in-language rather than translated. By script, 67% is Latin (English and romanized code-mix) and 33% native Indic (Tamil, Kannada, Telugu ≈ 9% each; Devanagari 5%; Bengali < 1%). Authoring in-language preserves the code-mixing and romanization real banking attacks use and a translation pipeline does not reproduce (Appendix G); the public supplement is the source of the Hindi contamination we detect and exclude (Appendix H).

Four design decisions distinguish BankShield from a general-purpose guard: (i) a cascaded architecture for on-premise deployment, (ii) direction-aware moderation that protects rather than polices the customer, (iii) lightweight retrainable probes, and (iv) a three-way policy separating blocking from human escalation. The biased-content detection of §6.2 follows from (ii) together with the adapted classifier and the trained discrimination probe (ablation in Appendix M).

5 Experimental Setup

Our evaluation is designed so that every reported figure is supported by released per-item logs or, where competitor logs were not retained, by frozen aggregate counts, and is robust to the decontamination and paired-metric requirements of §7; public baselines can be rerun, while BankShield's own inference cannot be regenerated end-to-end without the proprietary LoRA weights and production policy rules.

5.1 Decontamination and neutrality

A guard evaluated on data resembling its training set can report misleadingly favourable numbers. Every evaluation set below is decontaminated against BankShield's training data: we remove any item that exactly matches a training example, and any whose word-level (whitespace-tokenized) Jaccard similarity to the nearest is ≥ 0.70 (Lee et al., 2022), catching paraphrase leakage rather than only verbatim overlap (counts in Appendix H).

5.2 Paired-metric protocol

For each system we report the over-block rate (benign items wrongly flagged unsafe) and recall (unsafe items correctly flagged) on paired benign and unsafe sets, never a detection figure alone, with Wilson 95% confidence intervals throughout since several subsets are small. Because the key comparisons are between systems scored on the same items, we assess significance with McNemar's exact (binomial) test on paired per-item predictions; for the small number of unpaired aggregate comparisons we use Fisher's exact test. We believe reporting detection and over-block together, with paired significance, should become standard practice for financial safety systems. For BankShield's three-way decision (§2.2) we apply one convention throughout: for *harmful-content catch*, a block or a safe-directed route counts as a positive (a route escalates the harm to a human rather than ignoring it); for *benign over-block*, only a block counts (routing is a designed escalation, not a refusal); routing is additionally reported separately, and block-only figures are given alongside where they differ (e.g. output-safety catch 95.5% block-or-route vs 93.9% block-only).

5.3 Datasets

We evaluate on both external and domain-specific data.

External calibration data. We use XSTest (Röttger et al., 2024), 250 benign-but-sensitive prompts paired with 200 genuinely unsafe contrasts, as a standard, language-neutral probe of exaggerated safety, independent of any system's training data.

Benign banking data. To measure over-blocking on legitimate requests, we use 200 benign English banking queries from Banking-77 (Casanueva et al., 2020), a public intent-classification corpus with no India-specific training lineage, and 74 author-constructed benign Hindi/Hinglish banking queries, written by bilingual annotators and decontaminated as in §5.1. These isolate the failure mode of §2.1: refusing ordinary banking language.

Native-script data. We include 40 benign native-script (Devanagari and Tamil) items to test whether calibration depends on script rather than content.

Fairness data. For the counterfactual audit (§6.2), we sample 1,056 items from BharatBBQ (Tomar et al., 2025), an Indian-context bias benchmark: from this pool we draw 185 benign identity-mention and 123 biased items, and construct 160 minimal-pair counterfactual sets over religion, caste, region, and gender in which only the protected attribute varies while content is held constant. The same matched content, in stereotype-consistent and stereotype-violating framings, supports the direction-asymmetry analysis of §6.2.

Attack data. To measure recall on genuine banking attacks, we use 89 adversarial banking prompts spanning injection, fraud/OTP solicitation, KYC bypass, money-laundering, account-takeover, and officer impersonation, in English (35), Hindi (27), and romanized Hindi (27), authored by bilingual annotators to reflect deployment attack typologies (no real customer messages or logs are used) and decontaminated per §5.1.

Full per-dataset and per-language item counts are given in Appendix I.

5.4 Systems compared

We compare six systems on identical decontaminated data: BankShield (12B, proposed); L3Cube-IndicGuard (Bramhecha et al., 2026) (4B, Indic-specific); Llama Guard 3 (Grattafiori et al., 2024) (8B) and WildGuard (Han et al., 2024) (7B), two general-purpose guards; and two un-adapted backbones (the `gemma-3-4b-it` base the Indic guard builds on, and the `gemma-4-12B-it` backbone BankShield adapts) so adaptation can be isolated from scale (§6.2). Each competitor runs with its own released inference code; Prompt-Guard (86M) was excluded as scope-mismatched (Appendix J). Consistent with the workshop's open-research policy, we release our evaluation code, prompts, decontaminated benchmark sets, and fairness minimal-pair templates, and withhold the production rule sets and model weights of a deployed financial guard. All systems use greedy decoding, parsed into the block / route / allow decision space per Appendix L. The banking-fairness and output-safety evaluations (§6.3–§6.4) extend this panel with three further deployed guards, AWS Bedrock Guardrails, NVIDIA NemoGuard, and Prompt-Guard-2 (86M, included here as a lightweight floor rather than a scope-matched guard), and use the current *Llama Guard 4* (12B) in place of the Llama Guard 3 (8B) of the attack-recall and BharatBBQ panels (Tables 4 and 6); we note the Llama version differs across the two evaluation panels.[3]

6 Results

Our results test three claims: (H1) BankShield is a competent guard, competitive with the strongest systems, traced largely to its open backbone (§D); (H2) it combines low counterfactual flip with biased-content detection, reported as two separate axes rather than a joint guarantee (§6.2); (H3) detection holds beyond English (Appendix G).

6.1 Language and domain competence

BankShield is a competent banking guard: it matches or exceeds the strongest available systems on in-domain attack recall and benign over-refusal, and the ablation attributes most of this competence to its open backbone, with adaptation adding targeted detection (full per-system and per-

[3] Llama Guard by panel: version 3 (8B) in Tables 4 and 6; version 4 (12B) in Tables 2 and 3, reflecting the current release when each panel was built.

Configuration	Biased-content catch
Un-adapted 12B_0 backbone	24%
+ LoRA classifier	43%
+ discrimination probe (full)	**66.7%**

Table 1: What adaptation adds, on the 123-item biased set (output mode). Near-invariance, attack recall, and most over-block behaviour are inherited from the backbone; the distinctive measured gain from adaptation is biased-content detection, which still does not achieve proxy-versus-actuarial separation (§6.3).

language breakdown in Appendix D). An external seven-benchmark panel (Table 5) confirms this: BankShield attains the highest harm recall of any system compared, at the highest over-block cost. We focus the main text on the fairness findings, where the systems diverge most.

What the adaptation adds, versus the backbone. Decomposing biased-content catch separates the backbone's contribution from the adaptation's (Table 1): the un-adapted 12B_0 backbone catches 24%, LoRA fine-tuning raises this to 43%, and adding the output-side discrimination probe reaches 66.7%. Near-invariance, attack recall, and much of the over-block behaviour are inherited from the backbone; the one distinctive measured gain from adaptation is biased-content detection. Even so, this gain does not solve proxy-versus-actuarial separation (§6.3), and 66.7% does not significantly exceed the smaller Indic guard's 56.9% (p=0.09). This addresses the natural question of whether the backbone plus a lightweight post-filter would suffice: the probe adds detection, but the full system still lacks the selectivity the banking-fairness test requires.

6.2 Fairness: does the verdict depend on identity?

We first audit fairness on the general BharatBBQ benchmark, scoring each guard on flip (verdict change under identity substitution, on 160 *English* minimal-pair sets), biased-content catch, and benign over-flag (the catch and over-flag sets span English and Devanagari Hindi, $\sim$54/46%; Table 6). The result that organizes the paper: invariance and detection are separable, and BankShield alone attains low flip together with high catch, reported as two separate-axis measurements (flip on the counterfactual sets, catch on the biased set) rather than joint per-example fairness. Detection is uneven by language: over-flag 4.0% (en) vs 5.9% (hi) and catch 81.6% (en) vs 42.6% (hi), a competence gap we flag in Limitations. Per-attribute detail is in Appendix E.

6.3 From general bias to banking decisions

BharatBBQ is general; the discriminatory loan denial motivating this paper is not among its items. We construct a banking-specific fairness benchmark (256 items, in English, released); the code-mixed stress our thesis concerns is carried by the output-safety set below (35% Hinglish) and the romanized attack set (§5.4), and extending the coded-fairness items to code-mix is future work. Protected grounds follow the RBI Charter (Reserve Bank of India, 2014) (religion, caste, gender, age, physical ability), extended with region—not an RBI-enumerated ground but a discrimination axis in BharatBBQ and Indian case law; coded items instantiate group-proxy vocabulary of a kind Indian courts and official guidance recognize as discriminatory: the Supreme Court's treatment of caste-based proxies in *Sukanya Shantha v. Union of India* (Supreme Court of India, 2024) (a prison-context judgment we draw on by analogy, not a banking ruling) and the Supreme Court *Handbook on Combating Gender Stereotypes* (Supreme Court of India, 2023), which we adapt to banking decisions. The benchmark has three tiers: 144 minimal-pair *flips* (attribute varied, decision fixed); 32 *explicit* catch items (denial attributed to identity); and 40 *coded* pairs (36 denial-type and 4 *selective-friction*, where the adverse action is added procedural screening rather than an outright denial). Each coded pair matches a biased message whose adverse decision rests on a group-proxy ("applicants of your background lack repayment discipline") with a control justifying the same decision by a documented fact ("more than two missed EMIs"). Items are decontaminated (max Jaccard 0.33); Catch counts a block or discrimination flag; the control false-flag rate counts a wrongly flagged control.

Table 2 yields three findings. First, on the *paired* metric (the discriminating test, requiring a system to flag proxy-discrimination *and* clear the matched legitimate decision), no system does well: the best, WildGuard, reaches 37.5%. Explicit discrimination is largely caught (BankShield 100%, others 23–65%); coded proxy-discrimination is where the field breaks down. Second, the failures are structured, not random. AWS and the three open guards are *proxy-blind*: they both-clear most pairs, never flagging the coded bias (paired 0–

System	expl.	coded	ctrl-FF	paired	flip
BankShield	**100**	97.2	75.0	20.0	5/48
WildGuard	57.7	47.2	5.6	**37.5**	13/48
AWS Guard.	65.4	2.8	0	2.5	2/48
NemoGuard	61.5	0	0	0	0/48
Llama Guard 4	23.1	5.6	0	5.0	6/48
PromptGuard-2	0	0	0	0	0/48

Table 2: Banking fairness benchmark, output mode, in %. *expl.* / *coded*: catch on explicit / coded-proxy discrimination (explicit over 26 items, after excluding 6 malformed templates; *coded* over the 36 denial-type items); *ctrl-FF*: control false-flag rate on the 36 legitimate controls (expert-validated; Appendix I); *paired* (n=40, the 36 denial-type plus 4 selective-friction pairs): fraction where the biased half is flagged *and* the control cleared; *flip*: sets (of 48) whose verdict changes across group substitution (lower is fairer). Interpretation in §6.3. Wilson 95% CIs in Appendix I.

System	catch	over-block	paired
Llama Guard 4	53.0	6.1	**48.5**
NemoGuard v3	50.0	3.0	47.0
WildGuard	37.9	1.5	37.9
BankShield	**95.5**	68.2	28.8
AWS Guard.	27.3	3.0	24.2
PromptGuard-2	9.1	6.1	3.0

Table 3: Output-safety benchmark, in %. *catch*: harmful replies flagged; *over-block*: benign controls wrongly flagged; *paired* (n=66): harmful flagged *and* control cleared. On generic output-safety harms, BankShield is not the strongest separator: it ranks fourth on paired accuracy, catching nearly everything (95.5%) but over-blocking benign controls (68.2%), whereas mature general-purpose safety models are better calibrated on these mainstream harms. Wilson 95% CIs in Appendix I.

5%). BankShield is the mirror image: it both-*flags* most pairs (paired 20.0%), catching the proxy language but firing equally on the legitimate control: it detects "adverse banking message," not discrimination specifically. The 40 legitimate controls are expert-validated (two independent experts accepted 40/40 and 39/40; Appendix I), so this 75.0% is a genuine control false-flag rate. A sensitivity sweep confirms this is not a calibration artifact: no threshold separates proxy from actuarial reasoning, because false-flags decompose across detectors that share adverse-tone features (Appendix F). These catch figures are not an artifact of the three-way decision: counting only *block* (not route) as a flag leaves explicit catch unchanged at 100% and coded catch at 94.4%, versus 97.2% when routing is included. Low flip here also carries an over-block cost: BankShield flags 59.0% (85/144) of the benign flip variants (WildGuard 32.6%)—uniform over-blocking, not selective fairness.

A proxy-detection / group-fairness tension. Third, the one partial separator pays for it. WildGuard reaches the best separation (37.5%) but is the *only* system with substantial identity-conditioned flipping (13/48 flip-sets; others $\leq$6), and its identity-firing is bimodal: concentrated in *communal* classes (religion, caste, region), while on gender it separates with *zero* identity-attention cost. The observed pattern suggests a tension where identity is communal, but gender shows it is not an architectural necessity (directional, six grounds). BankShield occupies the opposite corner: even-handed (5/48) but unable to separate proxy from actuarial. No system achieves both.

6.4 Does the guardrail work in both directions?

BankShield claims *direction-aware* moderation (§4.3): it should catch harmful *responses*, not only inputs, which our input-side evaluation does not test. We build an output-safety benchmark of 66 minimal pairs (132 items, released): each pairs a harmful assistant reply (PII leakage, fraud-enabling, unsafe advice, abusive tone, jailbreak compliance, or mishandling a distressed customer) with a matched-benign control handling the *same* topic correctly. Items are detection targets, not instructions; self-harm items contain no method detail (43 English, 23 Hinglish). Discrimination is excluded (§6.3).

Table 3 is the mirror image of Table 2, reported as-is: BankShield's blanket strictness (fourth of six on paired accuracy) is a genuine model/calibration cost. Policy can change the operational consequence of a flag—block, route, or audit-log—but it does not improve the detector's ability to distinguish harmful from legitimate content (§6.5). One category resists the whole field: on mishandling a distressed customer, NemoGuard, AWS, and PromptGuard score *zero* and Llama Guard 4 only 2/11—a capability nearly absent from current tooling, with real stakes (per-category detail, Appendix C).

Guardrail competence is domain-specific. Tables 2 and 3 together make a point neither makes alone: on coded banking discrimination the general models are near-blind and the domain-tuned guard competitive, while on generic output-safety those models are well-calibrated and the domain-tuned guard over-blocks. Neither "general safety LLM" nor "domain-specialized guard" dominates.

6.5 Policy flexibility without retraining

BankShield's strictness is a policy setting, not a fixed property: on a fixed 100-item slice, applying three tenant policies (strict, deployed-default, permissive-advisory) to the *same* frozen model and probes with no retraining moves 84 of 100 items from block-first to monitor-first, so the same detections can be routed or audit-logged rather than blocked (per-policy action counts in Appendix M). The cost is explicit: the advisory policy fully allows two items it arguably should flag, a safety price we report rather than hide. This flexibility is at the action layer only: routing or audit-logging a flag changes its operational *consequence* (block, route, or log), not the detector's ability to tell discriminatory proxy reasoning from legitimate actuarial reasoning. The 75.0% control false-flag rate of §6.3 therefore remains a model and calibration failure even under a non-blocking policy, and the 84-of-100 action change is action-layer flexibility, not a fairness or classifier improvement.

7 Discussion

Fairness is not invariance; it is invariance without blindness. The general guards are counterfactually invariant yet detect little biased content (6–24%): their "fairness" is insensitivity, a guard blind to a discriminatory denial being no protection. Our backbone comparison locates the split, invariance from scale and detection from training, so fairness should be reported alongside detection and over-blocking, not assumed from invariance alone.

Paired evaluation and three lessons. A recall figure without its paired over-block rate cannot distinguish a calibrated guard from a permissive one; GUARDAUDIT's requirements (paired reporting, decontamination, a fairness audit) are model-, language-, and domain-agnostic. The banking findings yield three lessons: report proxy-versus-actuarial separation, not catch rate alone; test and report proxy-detection and group-fairness together, since our results suggest a possible trade-off; and evaluate guards per-domain.

8 Conclusion

We asked how to *measure* whether a safety guard for Indian-language banking is demographically fair, and offer GUARDAUDIT and two banking benchmarks as the answer. The evaluation shows invariance and detection are separable, and exposes a structured failure the field shares: no guard reliably separates coded proxy-discrimination from a matched legitimate denial (best paired accuracy 37.5%). BankShield illustrates this as a case study, not a solution: 0% flip and 66.7% catch (measured on separate sets, not a joint guarantee) alongside only 20.0% paired accuracy and a 75.0% false-flag rate on expert-validated legitimate controls, which leaves it not deployment-ready for autonomous discrimination-sensitive decisions. Guard evaluation must therefore measure detection and equitable treatment together, on matched data.

Limitations

Over-blocking, separate-axis fairness, and non-deployment-readiness. Two limitations compound. First, we do not lead on over-blocking: on XSTest-safe, Llama Guard 3 (3%) and WildGuard (2%) over-block far less than BankShield (18%), so our contribution is competitive detection and an audit protocol, not field-best calibration. Second, our fairness result is separate-axis: flip is measured on benign counterfactual pairs and catch on a separate biased set, so we do *not* claim that BankShield allows a benign identity-mentioning decision *and* flags its discriminatory counterpart on the *same* example. The closest same-content test we have is the 40 matched proxy/control pairs of Table 2, which pair a discriminatory rationale with a matched actuarial one for the same adverse decision; there BankShield reaches only 20.0% paired accuracy and over-flags 75.0% of expert-validated legitimate controls. Even those pairs do not additionally cross each rationale with demographic substitutions, so a full joint counterfactual evaluation (benign and biased variants of the same decision with identity swaps, scored together) remains future work. Taken together, these results mean BankShield is a useful detector but is *not* ready for autonomous use on discrimination-sensitive adverse decisions, where it must sit behind a human reviewer. Its 66.7% catch is not significantly above the Indic guard's (56.9%, p=0.09), so its distinction from that guard is its 0% flip against the guard's 19%, not detection.

Small samples. Several buckets are small (the Hindi/Hinglish and native-script banking subsets; the coded proxy-discrimination result rests on 40 minimal pairs across six grounds), with wide CIs; these results are directional, the proxy/group-fairness correlation is directional only, and the architectural reading of the confound would need a larger benchmark to establish precisely. We frame these as characterized observations, not proven laws, and, given the number of paired McNemar/Fisher tests on these small sets, report uncorrected p-values descriptively rather than as confirmatory.

Cross-language coverage and contamination. The standard Hindi hate-speech benchmark overlapped our training data at 98% (117/120 items), so we excluded it as memorized (Appendix G). On held-out data BankShield's Tamil hate detection (84%) far exceeds the general guards (2–18%) but trails its English performance; we evaluate two Dravidian languages (Tamil and Telugu, 72%), and broader low-resource coverage is future work.

Reproduction caveat. For L3Cube-IndicGuard, our rerun of its released model and inference code yields 46% over-blocking on XSTest-safe versus the accompanying work's reported 0.00% over-refusal. The definitions may differ, and we do not adjudicate the discrepancy.

Author-constructed and synthetic data. The banking-fairness, output-safety, 89-prompt attack, and 74-item benign banking sets were author-written, though to typologies externally grounded in the RBI Charter and proxy vocabulary from Indian case law (§6.3); the fairness headline uses external BharatBBQ items. Two independent bilingual annotators re-labeled a 159-item sample blind, agreeing at Cohen's $\kappa = 0.97$ (author agreement $\kappa = 0.94$–0.96) and surfacing three mislabeled attack items, since corrected; residual authorship circularity remains. Two independent experts validated all 40 legitimate controls behind the 75.0% control false-flag rate (accepting 40/40 and 39/40; Appendix I), so it is over-flagging on expert-validated controls; label validity remains construct-sensitive on the discriminatory side, where a stricter construct (an initial reviewer, $\kappa = 0.03$) would flag more. The corpus is 99.6% synthetic (a privacy choice; §4.7), so neither training nor evaluation need match real attacker distributions, and the only non-synthetic operational evidence is the 283-event volunteer pilot, which involved volunteers rather than real customers. Together, the 99.6% synthetic training corpus and the 283-event volunteer pilot provide little real-world validation; an externally-sourced or red-teamed evaluation set, and a study with real users, are future work.

Output-safety scope. The output-safety benchmark (§6.4) is 66 minimal pairs over six harm categories (English and Hinglish); it excludes discrimination and is not exhaustive. BankShield's fourth-of-six ranking there is a genuine calibration gap versus purpose-built safety LLMs, partly but not wholly addressable by policy (§6.5); we report it as measured. The benign distress controls originally cited a superseded crisis helpline, now corrected to Tele-MANAS (14416) in the released benchmark; re-running BankShield on the corrected controls marginally lowers its over-block (65.2% vs 68.2%), and the competitor systems could not be re-run, so Table 3 keeps the original all-systems run.

What the backbone supplies versus the training. The un-adapted `gemma-4-12B-it` backbone closely tracks BankShield on invariance, over-blocking, and attack recall (§6.2, Appendix M); much of a deployed guard's competence is thus the backbone's, and BankShield's contribution lies in the fairness-detection layer and compliant deployment, not in re-establishing baseline competence. Our conclusions are also Gemma-based and may not transfer to other backbone families.

Single domain and language coverage. BankShield is evaluated in one domain (banking) and one deployment context, and its language coverage is partial: hate-speech detection is evaluated in English, Tamil, and Telugu, while Kannada ($\sim$9% of training) is evaluated only for over-blocking, since its public unsafe benchmark is training-contaminated (Appendix G); Bengali ($<$1%) is unevaluated, and Marathi, Gujarati, and Punjabi are absent. Whether its calibration and fairness transfer to other settings, institutions, or these languages remains untested; broader coverage and intersectional identities are future work. Detection is also uneven within evaluated languages (English 81.6% vs Devanagari Hindi

42.6% on biased content; §6.2): invariant to identity but a weaker non-English detector, which is a priority to strengthen. The counterfactual flip sets are English-only, so the 0% flip is established for English; extending the minimal-pair audit to Devanagari Hindi is future work.

Ethics Statement

BankShield is a protective guardrail, not a censorship mechanism: its three-way action model blocks financial attacks and routes distress and grievances to human agents, so vulnerable customers are escalated to help rather than refused, supporting rather than replacing human decisions.

Data and privacy. Consistent with the constraints of §2.2, training and evaluation avoid real customer data; the system is built on synthetic and public data and self-hosted so customer data is not sent to external services.

Harmful content. Evaluating a safety system requires exposure to fraud, abuse, and other harmful content, handled only for building and testing protective measures; where human judgment was used (e.g. validating labels), we were mindful of annotator exposure to distressing material.

Annotators and pilot participants. The two independent annotators (§5) and the pilot testers were volunteers, former colleagues of an author, who participated with informed consent and without compensation. The volunteer test pilot did not serve real banking customers: its 283 events are these volunteers' own test interactions, so no real customer data was collected or retained.

Fairness. A guard that systematically treats demographic groups unequally can itself cause harm; our audit (§6.2) is motivated by this, and we report residual disparities honestly rather than only favourable measures, so deployers can decide informed.

Dual use and release. To avoid an adversarial roadmap and respect the security constraints of a financial system, we release our evaluation methodology and artifacts but not production attack-detection rules or model weights, for which the workshop organizers granted a commercial/security exception in writing.

References

Akshit Achara and Anshuman Chhabra. 2025. Watching the AI watchdogs: A fairness and robustness analysis of AI safety moderation classifiers. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 253–264.

Muhammad Farid Adilazuarda, Sagnik Mukherjee, Pradhyumna Lavania, Siddhant Singh, Alham Fikri Aji, Jacki O'Neill, Ashutosh Modi, and Monojit Choudhury. 2024. Towards measuring and modeling culture in LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.

Darpan Aswal and Siddharth D. Jaiswal. 2025. Phonetic perturbations reveal tokenizer-rooted safety gaps in LLMs. *arXiv preprint arXiv:2505.14226*.

Parth Bramhecha, Smit Deshmukh, Sairaj Bodhale, Adwait Borate, and Raviraj Joshi. 2026. IndicGuard: A multilingual safety guard model and dataset for Indic languages. *arXiv preprint arXiv:2606.22841*.

Iñigo Casanueva, Tadas Temčinas, Daniela Gerz, Matthew Henderson, and Ivan Vulić. 2020. Efficient intent detection with dual sentence encoders. In *Proceedings of the 2nd Workshop on NLP for ConvAI*.

Xin Jie Chua, Jeraelyn Ming Li Tan, Jia Xuan Tan, Soon Chang Poh, Yi Xian Goh, Debbie Hui Tian Choong, Foong Chee Mun, Sze Jue Yang, and Chee Seng Chan. 2025. Banking done right: Redefining retail banking with language-centric AI. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 646–658, Suzhou, China. Association for Computational Linguistics.

Thomas Davidson, Dana Warmsley, Michael Macy, and Ingmar Weber. 2017. Automated hate speech detection and the problem of offensive language. In *Proceedings of the 11th International AAAI Conference on Web and Social Media (ICWSM)*, pages 512–515.

Lucas Dixon, John Li, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2018. Measuring and mitigating unintended bias in text classification. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*.

Huaixia Dou, Jie Zhu, Minghao Wu, Shuo Jiang, Junhui Li, Lifan Guo, Feng Chen, and Chi Zhang. 2026. FinGuard: Detecting financial regulatory noncompliance in LLM interactions. *arXiv preprint arXiv:2605.29427*.

Sahaj Garg, Vincent Perot, Nicole Limtiaco, Ankur Taly, Ed H Chi, and Alex Beutel. 2019. Counterfactual fairness in text classification through robustness. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*.

Shaona Ghosh, Prasoon Varshney, Erick Galinkin, and Christopher Parisien. 2024. Aegis: Online adaptive AI content safety moderation with ensemble of LLM experts. *arXiv preprint arXiv:2404.05993*.

Government of India, Ministry of Electronics and Information Technology. 2023. The digital personal data protection act, 2023. Act No. 22 of 2023, https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf.

Aaron Grattafiori et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Seungju Han, Kavel Rao, Allyson Ettinger, Liwei Jiang, Bill Yuchen Lin, Nathan Lambert, Yejin Choi, and Nouha Dziri. 2024. WildGuard: Open one-stop moderation tools for safety risks, jailbreaks, and refusals of LLMs. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024) Datasets and Benchmarks Track*.

Yutao Hou, Yihan Jiang, Yuhan Xie, Jian Yang, Liwen Zhang, Hailiang Huang, Guanhua Chen, and Yun Chen. 2026. FinSafetyBench: Evaluating LLM safety in real-world financial scenarios. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 14181–14208, San Diego, California, United States. Association for Computational Linguistics.

Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabsa. 2023. Llama Guard: LLM-based input-output safeguard for human-AI conversations. *arXiv preprint arXiv:2312.06674*.

Jiaming Ji, Mickel Liu, Josef Dai, Xuehai Pan, Chi Zhang, Ce Bian, Boyuan Chen, Ruiyang Sun, Yizhou Wang, and Yaodong Yang. 2023. BeaverTails: Towards improved safety alignment of LLM via a human-preference dataset. *Advances in Neural Information Processing Systems*, 36.

Shijia Jiang, Yongfu Dai, Haochen Jia, Yuxin Wang, and Hao Wang. 2025. IntelliChain stars at the regulations challenge task: A large language model for financial regulation. In *Proceedings of the Joint Workshop of the 9th Financial Technology and Natural Language Processing (FinNLP), the 6th Financial Narrative Processing (FNP), and the 1st Workshop on Large Language Models for Finance and Legal (LLMFinLegal)*, pages 371–384, Abu Dhabi, UAE. Association for Computational Linguistics.

Raviraj Bhuminand Joshi, Rakesh Paul, Kanishk Singla, Anusha Kamath, Michael Evans, Katherine Luna, Shaona Ghosh, Utkarsh Vaidya, Eileen Margaret Peters Long, Sanjay Singh Chauhan, et al. 2025. CultureGuard: Towards culturally-aware dataset and guard model for multilingual safety applications. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*.

Khyati Khandelwal, Manuel Tonneau, Andrew M. Bean, Hannah Rose Kirk, and Scott A. Hale. 2024. Indian-BhED: A dataset for measuring India-centric biases in large language models. In *Proceedings of the 2024 International Conference on Information Technology for Social Good (GoodIT)*.

krishan-CSE. 2024. Tamil hate speech dataset. Hugging Face dataset, https://huggingface.co/datasets/krishan-CSE/Tamil_Hate_Speech.

Priyanshu Kumar, Devansh Jain, Akhila Yerukola, Liwei Jiang, Himanshu Beniwal, Thomas Hartvigsen, and Maarten Sap. 2025. PolyGuard: A multilingual safety moderation tool for 17 languages. In *Proceedings of the Conference on Language Modeling (COLM)*. ArXiv:2504.04377.

Matt J Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. 2017. Counterfactual fairness. In *Advances in Neural Information Processing Systems*, volume 30.

Katherine Lee, Daphne Ippolito, Andrew Nystrom, Chiyuan Zhang, Douglas Eck, Chris Callison-Burch, and Nicholas Carlini. 2022. Deduplicating training data makes language models better. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*.

Mounika Marreddy, Subba Reddy Oota, Lakshmi Sireesha Vakada, Venkata Charan Chinni, and Radhika Mamidi. 2022. Multi-task text classification using graph convolutional networks for large-scale low resource language. *arXiv preprint arXiv:2205.01204*. Telugu hate-speech data via Hugging Face dataset mounikaiiith/Telugu-Hatespeech.

Janki Atul Nawale, Mohammed Safi Ur Rahman Khan, Janani D, Mansi Gupta, Danish Pruthi, and Mitesh M. Khapra. 2025. FairI tales: Evaluation of fairness in Indian contexts with a focus on bias and stereotypes. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 30331–30380.

Vinodkumar Prabhakaran, Ben Hutchinson, and Margaret Mitchell. 2019. Perturbation sensitivity analysis to detect unintended model biases. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Qwen Team. 2025. Qwen3Guard technical report. arXiv:2510.14276.

M. Rajeshwar Rao. 2024. Innovations in banking – the emerging role for technology and AI. Reserve Bank of India Bulletin (speech delivered 22 December 2023; Bulletin dated 18 January

2024), https://rbi.org.in/scripts/BS_ViewBulletin.aspx?Id=22318.

Reserve Bank of India. 2014. Charter of customer rights. Reserve Bank of India (released 3 December 2014), https://rbidocs.rbi.org.in/rdocs/content/pdfs/CCSR03122014_1.pdf. Right to Fair Treatment; non-discrimination on grounds of gender, age, religion, caste, and physical ability.

Reserve Bank of India. 2018. Storage of payment system data. Reserve Bank of India, directive DPSS.CO.OD.No.2785/06.08.005/2017-2018; FAQs, https://www.rbi.org.in/commonman/english/scripts/FAQs.aspx?Id=2995.

Reserve Bank of India. 2023. Master direction on outsourcing of information technology services. Reserve Bank of India, https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=12486.

Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2024. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*.

Nihar Sahoo, Pranamya Kulkarni, Arif Ahmad, Tanu Goyal, Narjis Asad, Aparna Garimella, and Pushpak Bhattacharyya. 2024. IndiBias: A benchmark dataset to measure social biases in language models for Indian context. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 8786–8806.

Wenliang Shan, Michael Fu, Rui Yang, and Chakkrit Tantithamthavorn. 2025. SEALGuard: Safeguarding the multilingual conversations in Southeast Asian languages for LLM software systems. In *Proceedings of the 2nd IEEE/ACM International Conference on AI-Powered Software (AIware)*. ArXiv:2507.08898.

Olivia Sturman, Aparna R Joshi, Bhaktipriya Radharapu, Piyush Kumar, and Renee Shelby. 2024. Debiasing text safety classifiers through a fairness-aware ensemble. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 199–214, Miami, Florida, US. Association for Computational Linguistics.

Supreme Court of India. 2023. Handbook on combating gender stereotypes. Supreme Court of India (launched 16 August 2023), https://cdnbbsr.s3waas.gov.in/s3ec0490f1f4972d133619a60c30f3559e/uploads/2024/01/2024012544.pdf.

Supreme Court of India. 2024. Sukanya Shantha v. Union of India. Supreme Court of India, W.P.(C) No. 1404/2023, Neutral Citation 2024 INSC 753 (decided 3 October 2024). On caste-based discrimination and proxy vocabulary (prison context); applied here to banking by analogy.

Aditya Tomar, Nihar Ranjan Sahoo, and Pushpak Bhattacharyya. 2025. BharatBBQ: A multilingual bias benchmark for question answering in the Indian context. *arXiv preprint arXiv:2508.07090*.

Keyi Wang, Jaisal Patel, Charlie Shen, Daniel Kim, Andy Zhu, Alex Lin, Luca Borella, Cailean Osborne, Matt White, Steve Yang, Kairong Xiao, and Xiao-Yang Liu. 2025. FinNLP-FNP-LLMFinLegal-2025 shared task: Regulations challenge. In *Proceedings of the Joint Workshop of the 9th Financial Technology and Natural Language Processing (FinNLP), the 6th Financial Narrative Processing (FNP), and the 1st Workshop on Large Language Models for Finance and Legal (LLMFinLegal)*, pages 363–370, Abu Dhabi, UAE. Association for Computational Linguistics.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

Wenjun Zeng, Yuchi Liu, Ryan Mullins, Ludovic Peran, Joe Fernandez, Hamza Harkous, Karthik Narasimhan, Drew Proud, Piyush Kumar, Bhaktipriya Radharapu, et al. 2024. ShieldGemma: Generative AI content moderation based on Gemma. *arXiv preprint arXiv:2407.21772*.

A System Architecture

Figure 1 shows the full pipeline of §4: deterministic normalization and the lexical filter, the LoRA-adapted classifier, the three lightweight probes, and the per-tenant policy engine mapping their outputs to the three-way action space of §4.5.

B The GuardAudit Protocol

GuardAudit evaluates a deployed guard in four steps, each with a reporting obligation.

1. Decontaminate. Remove from every evaluation set any item that exactly matches a training example (on normalized text) and any whose word-level Jaccard similarity to its nearest training neighbor is ≥ 0.70; report per-set removal counts, and score all systems on the same decontaminated sets (§5.1, Appendix H).

2. Pair every detection figure. Report recall on unsafe items together with the over-block rate on matched benign items, never a detection figure alone, with Wilson 95% CIs; use McNemar's

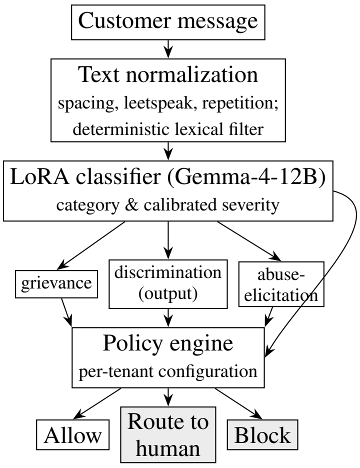

Figure 1: BankShield pipeline. A customer message is normalized, scored by a LoRA-adapted classifier, three lightweight probes, and a deterministic lexical filter, and mapped by a per-tenant policy engine to one of three actions: allow, route to a human, or block.

exact test for paired per-item comparisons and Fisher's exact test for unpaired aggregates (§5.2).

3. Audit the guard's own fairness. Construct counterfactual minimal pairs in which only a protected attribute varies, and report the three metrics together as a panel: flip rate (invariance), over-flag on benign identity mentions, and catch on biased content, so that invariance cannot be satisfied by blindness (§6.2). These are reported on their respective sets, not as a single same-item joint score.

4. Evaluate the deployed configuration. Score the action space the system actually deploys (here allow / route / block), counting designed escalations separately from refusals, and report policy sensitivity where thresholds are tunable (§4.5, §6.5).

The protocol is model-, language-, and domain-agnostic: steps 1–2 require access to the guard's training data, step 3 a bias benchmark for the deployment locale, and step 4 only the system's action space.

C Output-Safety Per-Category Detail

Table 3 is, deliberately, the mirror image of Table 2, and we report it as-is. On generic output-safety, BankShield ranks *fourth* of six on paired accuracy (28.8%): it catches nearly every harmful reply (95.5%) but over-blocks the benign controls (68.2%), while purpose-built safety LLMs, Llama Guard 4 (48.5%) and NemoGuard (47.0%), are better calibrated because these main-

Metric	BankSh.	$12B_0$	Indic	Base	LG-3	WildG.
Banking-attack rec. ↑	97	98	88	94	94	54
XSTest-unsafe rec. ↑	98	96	89	95	83	90
XSTest-safe o-block ↓	18	21	46	27	3	2
Banking o-block ↓	0	1	19	4	3	1

Table 4: Detection and over-block across six systems (four guards, the un-adapted `gemma-4-12B-it` backbone $12B_0$, and the un-adapted `gemma-3-4b-it` Base), neutral data, in %. Higher recall and lower over-block are better. The un-adapted 12B backbone tracks BankShield closely on banking-attack recall (98 vs 97), XSTest-unsafe recall (96 vs 98), and XSTest-safe over-block (21 vs 18), locating that competence largely in the backbone (§6.2). All systems except WildGuard catch 88–98% of Indic banking attacks. Banking o-block for BankShield counts block only (with routing counted as refusal it is 16%); the Indic guard's banking figure macro-averages its English (12%, n=200) and Hindi/Hinglish (26%, n=74) subsets. Competitor rows are the authors' runs; per-item logs were not retained. The released package is analysis-only (it regenerates tables from the released logs and includes the rival adapter code for reference); it is not a turnkey execution harness. Wilson 95% CIs in Table 8.

stream harms are squarely in their training distribution. BankShield's blanket strictness is a genuine model/calibration cost. Policy can change the operational consequence of a flag—block, route, or audit-log—but it does not improve the detector's ability to distinguish harmful from legitimate content (§4.6); we do not present it otherwise. One category resists the whole field: on mishandling a distressed customer, NemoGuard, AWS, and PromptGuard score *zero*, and Llama Guard 4 only 2/11, a guard that catches a banking assistant mishandling a customer in distress is nearly absent from current safety tooling, a gap with real stakes in a domain where such conversations occur.

D Competence and Ablation Detail

The fairness result of §6.2 is meaningful only if BankShield is a competent guard, that is, if it detects the banking attacks it is deployed to stop. We establish that here, against the strongest available guards on neutral data.

Banking-attack recall. On a neutral set of Indic banking attacks (n=89; English, Hindi, romanized Hindi), BankShield detects 97%, matching Llama Guard 3 (94%) and the un-adapted backbone (98%); even the small `gemma-3-4b-it` base (94%) and L3Cube (88%) are competent, while only WildGuard collapses on Indic input

(54%, $p{=}8{\times}10^{-12}$). Indic banking-attack competence is thus largely a property of a capable backbone, not specialized training: all systems except WildGuard catch 88–98%. What distinguishes guards is not this competence but fairness (§6.2).

On XSTest's unsafe set (external), BankShield reaches 98%, highest among the guards but close to the un-adapted backbone's 96% (Table 4), again largely the backbone's.

On over-blocking, we do not lead. On XSTest-safe, Llama Guard 3 (3%) and WildGuard (2%) over-block far less than BankShield (18%). BankShield's 18% closely tracks the un-adapted `gemma-4-12B-it` backbone (21%), so this calibration (and the low benign-banking over-block, 1% un-adapted, ≈0% for BankShield) is the backbone's, not our adaptation's.

The Indic guard's reported 0.00% over-refusal rate does not reproduce: with its released model and code we observe 46% over-blocking on XSTest-safe (Appendix K). We draw only the methodological point: a near-zero over-refusal figure is uninformative unless reported with unsafe-set recall on the same data.

The over-block rates above count only benign content that is blocked (a hard refusal that denies a customer legitimate service), not content routed to a human (a designed escalation for grievance or distress that preserves continuity of service); counting routing as refusal would raise the banking figure to 16%, but such routing is the intended behaviour, not an error.

E General-Fairness Per-Attribute Detail

This is the central question of our evaluation. Prior Indic-safety work treats identity as a category of harm to detect; we instead ask whether the guard is itself biased: whether its verdict changes when only a protected attribute (caste, religion, region, gender) varies, silently over-policing some groups. We audit all six systems on counterfactual minimal pairs extending BharatBBQ (Tomar et al., 2025). We report three metrics, all on the same output-side framing (the assistant-facing text) so no system is advantaged by our direction-aware design. *Flip* is the fraction of the 160 minimal-pair sets whose verdict changes when only the protected attribute is swapped (content fixed); *over-flag*, the fraction of 185 benign identity-mention items wrongly flagged (counting a block or discrimination-flag, not routing); *catch*, the fraction of 123 biased items correctly flagged. Because BankShield's discrimination detection is output-side by design (§4.3), catch spans operating points. For BankShield, catch is the output-side action catch: of the 82 caught items all are blocked and none routed, so block-only catch equals block-or-route catch (66.7%); the discrimination probe alone flags 50.4% (62/123), the remaining catches arising from the other output-side detectors. Per-item actions are released.

Table 6 reveals a clear divide. On flip, BankShield is invariant (0.0%), as are both general guards (0%), while the Indic guard flips on 19% and the base on 16% (McNemar vs BankShield $p{=}9{\times}10^{-10}$, $6{\times}10^{-8}$). Flip must be read against the any-flag rate on these benign pairs (lower is better): a set can only flip if a guard flags at least one variant, and flagging benign identity content is itself undesirable. BankShield flags none of the 160 benign sets (as do the general guards), the correct behaviour on benign content, whereas the Indic guard and the base flag many (38 and 25) and then treat the groups differently, flipping on 82–100% of those. Among guards that leave benign pairs alone, what distinguishes BankShield is that it *also* detects biased content: the general guards detect little (Llama Guard 3 6%, WildGuard 24%; both $p{<}10^{-14}$), while BankShield catches 66.7% on the separate 123-item biased set at a 4.9% over-flag (block-only, per §L; 5.4% counting routing). Its catch does not significantly exceed the Indic guard's 56.9% ($p{=}0.09$). Taken together, 0% flip on the benign counterfactuals and 66.7% catch on the biased set rule out simple flag-nothing behaviour, but, measured on different item sets, do not establish joint per-example fairness.

The general guards' invariance is uninformative: they detect little of the biased content a guard must catch, so a guard that cannot recognize a discriminatory loan denial offers no protection. BankShield stays invariant while retaining the ability to detect biased content. To locate the source, we evaluate the un-adapted `gemma-4-12B-it` backbone on the same pairs: it flips on 2%, statistically indistinguishable from BankShield's 0% ($p{=}0.25$), whereas the smaller `gemma-3-4b-it` base flips on 16%. The invariance is thus a property of backbone scale, not our adaptation. What training adds is detection: an ablation traces the biased-content catch from the un-adapted backbone's 24%, to 43% with the LoRA-adapted classifier (+19), to 66.7% with the added discrimination probe (+24; Appendix M).

Boundaries of the result. BankShield over-flags benign identity mentions at 4.9% (block-only; 5.4% if routing is counted, still above the general guards' 0–1%, the price of not being insensitive to identity), catches biased content at 66.7% (output) and 47% (input), and retains a mild 5-point asymmetry across bias directions (stereotype-consistent vs stereotype-violating framings). We do not claim joint per-example fairness, since flip and catch are measured on separate item sets; the defensible reading is that BankShield leaves benign identity content unflagged (0% flip) while detecting biased content the general guards miss.

F Sensitivity Sweep Detail

Sweeping the discrimination threshold (Low/Med/High) over the 40 coded pairs trades catch against false-flag monotonically and never separates: BankShield output paired 15→7→4 of 40, input flat at 19–20 of 40 at every setting; AWS 0→1→1. At the least-sensitive setting the eight residual control false-flags decompose as threat 3, dangerous 1, abuse 2, AML 1, injection 1, with *none* from the discrimination probe, so even a perfect discrimination threshold cannot separate. WildGuard exposes no sensitivity knob. This isolates the confound to the architecture (adverse-tone across detectors), not calibration.

G Cross-Language Detection Detail

The subsets (English 120, Tamil 160) were drawn by uniform random sampling with fixed seeds after decontamination; for Davidson, only hate-labeled items are the unsafe class (offensive excluded); the Tamil subset uses the corpus hate label, deduplicated.

A second question is whether BankShield's detection holds across Indic languages. We test on held-out hate-speech benchmarks in English (Davidson et al., 2017) and Tamil (krishan-CSE, 2024), decontaminated against our training corpus. These probe cross-script rather than code-mixed competence; our romanized and code-mixed evaluation is concentrated in the banking-attack set (§5), and extending code-mixed hate-speech evaluation to the Dravidian languages is future work. (We excluded Hindi: the standard benchmark overlapped our training data, which we report in Limitations rather than present as a memorized result.) The pattern is stark. On English hate (Davidson, n=120) the general guards are not weak: WildGuard is in fact the strongest of the three at 98% [93–99] versus BankShield's 82% [75–88] and Llama Guard 3's 65% [56–73]. But on Tamil (n=160) their detection drops sharply: BankShield reaches 84% [78–89] against Llama Guard 3's 18% [13–25] and WildGuard's 2% [1–6] ($p<10^{-30}$). The general guards are English-specialized; BankShield stays competent across scripts.

We add a second Dravidian language, Telugu, on a held-out public hate benchmark (Marreddy et al., 2022) decontaminated against our corpus: BankShield catches 72% (55/76) of native-script Telugu hate, a second point of Dravidian competence, at a higher benign over-block (29% of 150) consistent with the looser non-English calibration of §6.2. For Kannada the picture is different: the public offensive benchmark overlapped our training data as heavily as the Hindi one (all native-script unsafe items matched), so we exclude it. A synthetic benign-only check finds a low over-block (5%, 4/79) on native Kannada, but a held-out Kannada *unsafe* evaluation, which neither the contaminated public data nor a safety-aligned generator can supply and which machine translation would violate our in-language principle (§4.7) to obtain, remains future work.

Cross-language competence does not itself confer fairness (the multilingual Indic guard remains identity-sensitive, 19% flip), so the two results are complementary: §6.2 shows fairness on identity swaps; this section shows detection does not depend on English.

Together the results support H1–H3: BankShield is competitive on banking detection, and combines low counterfactual flip with biased-content detection (reported as separate axes, not a joint guarantee), while extending detection to other scripts, though unevenly (Tamil 84%, Telugu 72%, but Devanagari Hindi 42.6% on biased content).

H Decontamination Details

All evaluation sets are decontaminated against 401,274 unique training and evaluation texts (525,094 rows across 14 files; a superset of the ≈172K SFT set of §4.7, also including the fairness training set and held-out evaluation material) by exact-match on normalized text plus word-level (whitespace-tokenized) Jaccard similarity $\geq$ 0.70; the deduplication script is released with the benchmark artifacts. Removal counts: author-constructed Hindi/Hinglish banking, 6 of 80 removed (74 retained); native-script benign, 0 of 40; Banking-77, 0 (exact and near); XSTest, 0; the fairness benchmark (BharatBBQ), 0 exact and 0 near-duplicate, including against both the 11,907-item fairness training set and the 4,499-item demographic recalibration set that together train the output-side discrimination probe (13,622 safe/discrimination items after label filtering), confirming the fairness result behind the 66.7% catch is held-out; native hate-speech, 0 overlap for English and Tamil, but 117 of 120 (98%) for the standard Hindi benchmark, which was consequently excluded (see Limitations). An independent rerun of the near-duplicate step over the released one-way-hashed training index reproduces these figures *approximately*, not exactly: it yields 1,978 total Jaccard-$\geq$0.70 near-duplicates against the public benchmarks versus the 1,961 in the reported per-set counts (a $\approx$1% difference attributable to index-rebuild ordering).

I Dataset Statistics

Evaluation sizes (full detail Appendix H): XSTest 250/200; Banking-77 200 benign; Hindi/Hinglish benign 74; native-script benign 40; BharatBBQ source pool 1,056, evaluated target 868 unique prompts (185 benign, 123 biased, and 560 variants in 160 minimal-pair sets over religion/caste/region/gender); 89 Indic banking attacks. Banking-77 has no India-specific training lineage; the author-constructed banking and attack sets were verified disjoint from training by the procedure of Appendix H.

All author-constructed sets were written by a single bilingual author with banking-domain familiarity; two independent bilingual annotators relabeled a 159-item stratified sample blind to the answer key, following a written protocol released with the artifacts (inter-annotator $\kappa = 0.97$; see Limitations for the residual authorship-circularity caveat).

Control validation. We validate the banking control labels in two blind studies. First, on a 120-item stratified sample, an initial reviewer applied a strict construct—treating any adverse decision that names a protected attribute as discriminatory regardless of the stated reason—accepting only 3 of 57 controls (Cohen's $\kappa = 0.03$), while a reviewer under clarified instructions (an objective documented reason makes an incidental protected-attribute mention legitimate) agreed with our labels ($\kappa = 0.93$); the disagreement is construct-driven. Second, to cover the exact tier behind the 75.0% control false-flag rate, two independent domain experts blind-labeled all 40 coded pairs (80 items: 40 discriminatory, 40 matched controls). Both accepted essentially every legitimate control (40/40 and 39/40), so the controls are expert-validated and the 75.0% is a genuine over-flag rate; they agreed with our labels at $\kappa = 0.80$ and 0.75 (expert–expert $\kappa = 0.90$), confirming 32/40 and 31/40 of the coded-discriminatory items (the coded-proxy tier is subtle, so some read as borderline—a construct-sensitivity we retain). Reviewers are volunteer bilingual professionals with banking or fairness-evaluation experience who worked independently and blind to the answer key from the written instructions released with the artifacts, with informed consent and no compensation; per-item ratings are released with identities anonymized.

Licenses. External datasets are used under their published licenses: XSTest, Banking-77, and the Telugu hate-speech corpus under CC-BY-4.0; the Davidson English hate-speech corpus under MIT; the Tamil hate-speech corpus under Apache-2.0; BharatBBQ under the MIT license declared on its official HuggingFace release (which covers the English and Hindi splits we use). Our author-constructed benchmarks contain no items copied from these sources (Appendix H); the released BharatBBQ-derived minimal-pair sets redistribute source items under that license with attribution.

Table 8 reports Wilson 95% confidence intervals for the point estimates in Tables 4 and 6.

Confidence intervals for the banking and output-safety tables. Wilson 95% intervals for BankShield's headline rates in Tables 2 and 3. Banking (§6.3): explicit catch 100% [87.1, 100], coded catch 97.2% [86, 100], control false-flag 75.0% [59, 86], paired 20.0% [10, 35]. Output-safety (§6.4): catch 95.5% [87.5, 98.4], over-block 68.2% [56.2, 78.2], paired 28.8% [19.3, 40.6]. On these n=26–66 sets the intervals are wide; we read the banking and output-safety point estimates as directional accordingly.

J Baseline Systems

All baselines are evaluated on the same decontaminated sets as BankShield and appear directly in Tables 4–6: L3Cube-IndicGuard (run with its released inference script), the un-adapted `gemma-3-4b-it` base model, Llama Guard 3 (8B), and WildGuard (7B). Prompt-Guard (86M) was evaluated but excluded from the tables as scope-mismatched: it is a jailbreak/injection detector and is inert on this content-safety evaluation (flagging none of the benign or the toxicity/violence content under its intended configuration), so reporting it alongside content-safety guards would mislead.

K Reproduction of a Reported Over-refusal Figure

The Indic guard's accompanying work (Bramhecha et al., 2026) reports a 0.00% over-refusal rate on XSTest safe prompts. Running its released model with its released inference script and prompt format on the XSTest-safe set, we observe a 46% over-block rate (95% CI [40–52]). Illustrative benign prompts it flags include innocuous questions rephrased with sensitive surface words. We do not adjudicate the discrepancy with the reported figure, which may reflect a different definition of over-refusal; we report only what the released artifacts produce, and use it solely to motivate reporting over-block together with unsafe-set recall on the same data.

L Output Parsing and Decision Mapping

BankShield uses greedy decoding (`do_sample=False`); its category and severity are obtained from a single forward pass by scoring log-probabilities over a fixed set of label tokens rather than by free-form generation, and its probes are sigmoid scores over sentence embeddings. All decoding is deterministic, so results are seed-independent and no sampling seed is involved. Comparison systems are run greedily with a 96-token generation budget and parsed from JSON by regular expression; on L3Cube-IndicGuard this yielded 0 parse errors over 764 items. For the decision mapping, a comparison system's unsafe label is treated as a flag; for BankShield, a block or route is treated as a flag, with over-block rates counting block only and routing reported separately. The un-adapted backbones ($12B_0$ and the 4B base) are scored with the same deterministic label-token procedure and instruction template as BankShield's classifier, differing only in the absence of the LoRA adapter and probes.

M Training and Deployment Details

Classifier adaptation. BankShield adapts `google/gemma-4-12B-it` with LoRA (rank r=16, α=32, dropout 0.05; target modules: the attention projections q, k, v, o and the MLP gate, up, down), served in 4-bit NF4 with double-quantization and bfloat16 compute. Training used AdamW with a learning rate of 2×10^{-4} on a cosine schedule (warmup ratio 0.03), one epoch over the ≈172K-example corpus, an effective batch size of 32 (8×4 gradient accumulation), and a maximum sequence length of 128 tokens, on a single NVIDIA A4000 (16 GB).

Probes. The three learned probes (grievance, output-side discrimination, abuse-elicitation) are logistic-regression heads over 768-dimensional `intfloat/multilingual-e5-base` embeddings (input formatted as `query:` + text, truncated to 192 tokens, mean-pooled and L2-normalized), trained with balanced class weights; a fourth signal is a deterministic normalized-lexicon match. Per-probe thresholds are stored in configuration and tunable per tenant. The output-side discrimination probe is trained on 13,622 safe/discrimination-labelled items (4,546 discrimination, 9,076 safe) drawn from two in-house synthetic sources: the fairness training set (§4.7) and a counterfactual-balanced demographic recalibration set. Both sources are verified disjoint from the BharatBBQ evaluation data (0 exact and 0 word-level Jaccard $\geq$ 0.70 near-duplicates across all 1,102 pooled prompt occurrences (934 unique; 868 target prompts after excluding 66 `author2` records); Appendix H), so the 66.7% catch is a held-out result. An ablation decomposes the biased-content catch (n=123, output mode) across components: the un-adapted `gemma-4-12B-it` backbone catches 24%; LoRA fine-tuning raises the classifier alone to 43% (+19); adding the output-side discrimination probe raises the full system to 66.7% (+24), recovering 29 cases the classifier misses. Counterfactual invariance, over-block, and attack recall are, by contrast, inherited from the backbone (§6.2), so fine-tuning's distinctive contribution is biased-content detection: the probe is its decisive component, though not its sole source.

Deployment status. BankShield runs as a self-hosted volunteer test pilot on a single on-premise NVIDIA A4000 (16 GB), serving a small number of volunteer testers (former colleagues of an author, not real banking customers); to date it has handled 283 test moderation events (snapshot 8 July 2026), with no real customers and no banking actions taken. The self-hosted, on-premise configuration is what satisfies the localization constraint of §2.2. This is an unoptimized pilot, not a production deployment: on this hardware end-to-end latency is p50 $\approx$ 0.7 s / p95 $\approx$ 3.2 s, which production use would reduce with a larger GPU and request batching. Any deployment language in this paper refers to this pilot and its regulatory constraints, not to production scale, and any scale described in §1 refers to the setting BankShield targets, not its current user base.

Per-policy action counts (§6.5). On the fixed 100-item slice, the same frozen model and probes under three tenant policies (no retraining) yield: strict, 96 blocks; deployed-default, block-first; permissive-advisory, monitor-first (39 route, 37 flag, 12 block, 12 allow). 84 of 100 items change action across policies, and the permissive-advisory policy fully allows 2 items that arguably warrant a flag.

Recall ↑	**PolyG**	**HarmB**	**HH**	**XST**	**JbB**	**ToxC**	**Pooled**
BankShield	95	100	75	98	100	90	**90.9** [89–92]
PolyGuard	62	92	75	88	96	81	74.1 [73–75]
WildGuard	36	99	78	91	98	74	73.5 [73–74]
AWS Bedrock	35	81	57	86	88	92	69.0 [66–72]
L3Cube-IndicGuard	61	94	69	89	92	49	67.6 [67–68]
Qwen3Guard	46	97	61	80	96	56	60.2 [59–61]
Llama Guard 3	43	97	45	79	94	32	45.2 [44–46]

Over-block ↓	**PolyG**	**HH**	**XST**	**JbB**	**ToxC**	**ORB**	**Pooled**
BankShield	46	2	18	65	14	16	20.7 [19–23]
PolyGuard	7	4	5	64	11	11	8.0 [8–8]
WildGuard	1	1	2	50	6	13	4.5 [4–5]
AWS Bedrock	24	1	23	38	15	17	16.4 [14–19]
L3Cube-IndicGuard	19	7	46	47	5	19	9.7 [9–10]
Qwen3Guard	1	0	0	14	1	4	1.1 [1–1]
Llama Guard 3	3	0	3	19	3	1	1.7 [2–2]

Table 5: External-benchmark competence panel: harm recall (top) and benign over-block (bottom), %, across seven public safety benchmarks, decontaminated per §5. *Pooled* is the set-size-weighted rate with Wilson 95% CI; per-set cells are point estimates (per-set counts released). **Protocol asymmetry, and its direction.** BankShield and AWS are scored on Jaccard-0.70–deduplicated stratified samples (250/class); the open-weight rivals on the full exact-deduplicated sets. The stricter deduplication is applied *only to our rows* and removes our nearest-to-training (hence easiest) items; we flag this sampling asymmetry as a caveat rather than claim it is strictly conservative, and note only that it does not inflate BankShield's rows. BankShield attains the highest harm recall (90.9%) at the highest over-block (20.7%); the low-over-block guards reach calibration by missing 40–55% of harm. JailbreakBench-safe is adversarial-benign and over-blocked by every system (47–65%). Llama Guard 4 is omitted: its Llama-4 backbone has unstable inference in current `transformers` releases, so Llama Guard 3 represents the family. The XSTest column (98/18) matches Table 4. Sampled-row per-set n (BankShield/AWS), recall: polyguard 250, harmbench 152, hh-rlhf 250, xstest 189, jailbreakbench 49, toxicchat 250; over-block: polyguard 250, hh-rlhf 250, xstest 207, jailbreakbench 34, toxicchat 250, or-bench 250. Open-weight rivals are scored on the full decontaminated sets; per-set counts (aggregate k/n, from which the Wilson CIs recompute) are released in `rivals_*.json`.

Metric	BankSh.	$12B_0$	Indic	Base	LG-3	WildG.
Flip ↓	0.0	2	19	16	0	0
Over-flag ↓	4.9	2	13	4	1	0
Catch biased ↑	66.7	24	56.9	33	6	24

Table 6: Guard fairness on Indian-context counterfactuals, in %. Arrows indicate the better direction. $12B_0$ is the un-adapted `gemma-4-12B-it` backbone: it is already near-invariant (flip 2%) but catches little biased content (24%), showing that invariance comes from backbone scale while detection (24% → 66.7%) is what adaptation adds. BankShield combines zero flip (0.0%) with the highest biased-content catch (66.7%, on the separate 123-item biased set); measured on different data, these rule out flag-nothing behaviour but do not establish joint per-example fairness. LG-3 (Llama Guard 3) and WildGuard are invariant only by catching almost nothing. All BankShield metrics here are computed in output mode: flip on 160 minimal-pair sets, over-flag on 185 benign items, catch on 123 biased items. BankShield's flip is significantly below the Indic guard's ($p<10^{-9}$) and its catch far exceeds both general guards' ($p<10^{-14}$), McNemar. Wilson 95% CIs in Table 8.

	Indic	Bank.	Fair.	Deploy.
Llama Guard 3 / ShieldGemma	part.	–	–	–
CultureGuard	yes	–	–	–
L3Cube-IndicGuard	yes	–	–	–
BankShield	yes	yes	yes	yes

Table 7: Where prior guards sit on the dimensions this paper addresses: Indic coverage, banking focus, a fairness audit *of the guard itself*, and evaluation in deployment (for BankShield, a self-hosted volunteer test pilot; §1). CultureGuard (NVIDIA, Llama-based) and L3Cube-IndicGuard (Gemma-based) are distinct systems; we evaluate the latter. ("Fair." = fairness audit.)

Metric (system)	n	%	[95% CI]
Banking-attack rec. (BankSh.)	89	97	[91–99]
Banking-attack rec. ($12B_0$)	89	98	[92–99]
Banking-attack rec. (Indic)	89	88	[79–93]
Banking-attack rec. (Base)	89	94	[88–98]
Banking-attack rec. (LG-3)	89	94	[88–98]
Banking-attack rec. (WildG.)	89	54	[44–64]
XSTest-unsafe rec. (BankSh.)	200	98.5	[96–99]
XSTest-unsafe rec. ($12B_0$)	200	96	[92–98]
XSTest-safe o-block (BankSh.)	250	18	[14–23]
XSTest-safe o-block (Indic)	250	46	[40–52]
XSTest-safe o-block ($12B_0$)	250	21	[16–26]
Flip (BankSh.)	160	0.0	[0–2.3]
Flip ($12B_0$)	160	2	[1–6]
Flip (Indic)	160	19	[14–26]
Flip (Base)	160	16	[11–22]
Over-flag (BankSh.)	185	4.9	[3–9]
Over-flag ($12B_0$)	185	2	[1–5]
Over-flag (Indic)	185	13	[9–19]
Catch biased (BankSh.)	123	66.7	[58–74]
Catch biased ($12B_0$)	123	24	[18–33]
Catch biased (Indic)	123	56.9	[48–66]
Catch biased (LG-3)	123	6	[3–12]
Catch biased (WildG.)	123	24	[17–32]

Table 8: Wilson 95% confidence intervals for the main point estimates in Tables 4–6. Percentages are exact where non-integer. Cross-language hate-speech CIs are given inline in Appendix G.